Discount Increase Analysis Q4 vs Q3 2025

Dataset: E-commerce orders · Generated: 2026-11-07 00:04 UTC

Question

Which product category saw the largest increase in average discount in Q4 2025 compared with Q3 2025?

Executive summary

The product category with the largest increase in average discount_pct between Q3 2025 and Q4 2025 is Electronics. Its average discount rose from 0.0808 in Q3 to 0.2211 in Q4, a delta of 0.1402 (173.5% increase). All other categories either declined slightly or remained essentially unchanged. This suggests a targeted discount strategy for Electronics during the fourth quarter of 2025.

Scope & task alignment

Scope	Value
Current period	Q4 2025
Comparison period	Q3 2025
Metric	discount_pct
Grouping	product_category
Task alignment	ALIGNED

Assumption: Periods are relative to the data, which covers 2024-01-01 to 2025-12-31 in 'order_date', not to today's date.

Run metric	Value
iterations	1
tool calls	3
sql queries	1
findings verified	3/3
numeric claims supported	10/10
hallucination rate	0
task alignment	aligned
total tokens	18,251
latency s	72.66

Findings

1. Electronics shows the biggest rise in average discount [VERIFIED]

Average discount_pct for Electronics increased from 0.0808 in Q3 2025 to 0.2211 in Q4 2025, a rise of 0.1402 (173.5% increase).

Numbers checked: 4/4 matched tool outputs. Evidence: E2, E1

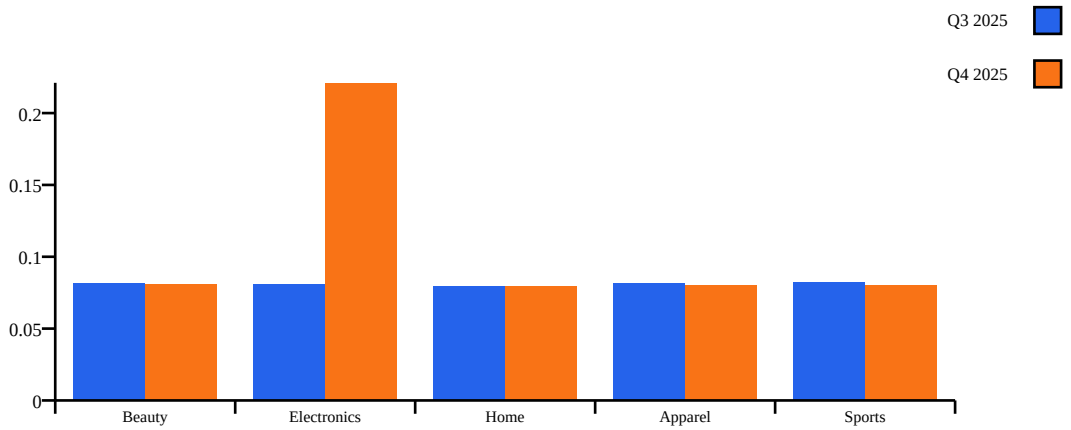
2. Other categories experienced minimal changes [VERIFIED]

Home, Beauty, Apparel, and Sports each had average discount changes between -0.0018 and -0.0004, indicating slight declines or negligible variation.

Numbers checked: 2/2 matched tool outputs. Evidence: E2

Visualizations

Average Discount_pct by Product Category: Q3 vs Q4 2025



Limitations & caveats

- The data only covers orders up to 2025-12-31, so Q4 2025 includes only three months of data.
- Average discount_pct may be affected by promotional campaigns that are not captured in the dataset.
- Small sample sizes for some categories could exaggerate percentage changes.

Supporting evidence

[E0] profile_dataset - Dataset understanding

18861 rows, 13 columns; 0 warnings

[E1] period_comparison -

```
SELECT "product_category"::VARCHAR AS "product_category", CASE WHEN "order_date" >= '2025-10-01'
AND "order_date" < '2026-01-01' THEN 'current' ELSE 'previous' END AS period, AVG("discount_pct")
AS value FROM "ecommerce_orders" WHERE (("order_date" >= '2025-07-01' AND "order_date" <
'2025-10-01') OR ("order_date" >= '2025-10-01' AND "order_date" < '2026-01-01')) GROUP BY ALL
```

product_category	previous	current	delta	pct_change	share_of_total_cha
Sports	0.08	0.08	0	-2.15	-1.3
Apparel	0.08	0.08	0	-1.53	-0.92
Beauty	0.08	0.08	0	-0.75	-0.45
Home	0.08	0.08	0	-0.57	-0.33
Electronics	0.08	0.22	0.14	173.54	103

[E2] run_sql - Get avg discount per category for Q3 and Q4 and compute differences

```
WITH q3 AS (
SELECT product_category, AVG(discount_pct) AS avg_discount_q3
FROM "ecommerce_orders"
WHERE order_date >= DATE '2025-07-01' AND order_date < DATE '2025-10-01'
GROUP BY product_category
), q4 AS (
SELECT product_category, AVG(discount_pct) AS avg_discount_q4
FROM "ecommerce_orders"
WHERE order_date >= DATE '2025-10-01' AND order_date < DATE '2026-01-01'
GROUP BY product_category
)
SELECT q3.product_category,
q3.avg_discount_q3,
q4.avg_discount_q4,
(q4.avg_discount_q4 - q3.avg_discount_q3) AS delta,
CASE WHEN q3.avg_discount_q3 = 0 THEN NULL ELSE (q4.avg_discount_q4 -
q3.avg_discount_q3)/q3.avg_discount_q3 END AS pct_change
FROM q3
JOIN q4 USING (product_category)
ORDER BY delta DESC;
```

product_category	avg_discount_q3	avg_discount_q4	delta	pct_change
Electronics	0.08	0.22	0.14	1.74
Home	0.08	0.08	-0	-0.01
Beauty	0.08	0.08	-0	-0.01
Apparel	0.08	0.08	-0	-0.02
Sports	0.08	0.08	-0	-0.02

Agent execution timeline

#	Type	Step	Duration	Status
1	node	understand_dataset	101 ms	ok
2	tool	profile_dataset	100 ms	ok
3	discovery	mcp_tool_discovery	0 ms	ok
4	node	resolve_time_scope	41 ms	ok

#	Type	Step	Duration	Status
5	decision	time_scope_resolved	0 ms	ok
6	node	plan	20531 ms	ok
7	llm	plan	20531 ms	ok
8	decision	task_contract	0 ms	ok
9	llm	act (iteration 1)	5121 ms	ok
10	tool	period_comparison	48 ms	ok
11	llm	act (iteration 1)	5851 ms	ok
12	tool	run_sql	47 ms	ok
13	llm	act (iteration 1)	8403 ms	ok
14	tool	create_chart	82 ms	ok
15	llm	act (iteration 1)	3598 ms	ok
16	node	reflect (iteration 1)	5184 ms	ok
17	llm	reflect	5184 ms	ok
18	decision	stop_investigation	0 ms	ok
19	node	synthesize	23582 ms	ok
20	llm	synthesize	23582 ms	ok
21	node	verify_findings	1 ms	ok
22	node	generate_report	0 ms	running